Anisotropic Casimir stress on rectangular flat T^3 and a no-attractor result for an instantaneous Bianchi-I closure

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Compact spatial topology can source anisotropic vacuum stress. For a free massless scalar on a rectangular flat three-torus we test an already-renormalized periodic lattice formula for Casimir energy density and directional pressures against cutoff convergence, cubic isotropy, thermodynamic pressure derivatives, conformal trace, dimensional scaling, and an independent heat-kernel/Mellin evaluator at four cubic and noncubic points. We then transport the static shape-dependent stress through a declared instantaneous constitutive closure with a^{-4} dilution, first as a perturbation of de Sitter expansion and then at finite amplitude on a biaxial Bianchi-I expanding branch with positive cosmological-background density. The latter reconstructs the mean expansion from the Hamiltonian constraint; it is not a fixed-de Sitter or uniformly perturbative calculation. In the bounded scan reported here, this closure produces only *transient* passages near the historically proposed ratio $H_t/H_p = 13/12$, with no quasi-plateau or late-time attractor. Within the registered domain, this excludes a maintained ratio for the tested closure and makes the target a discriminator between instantaneous free-field stress and any future driven or memory-bearing constitutive law. It is not a general no-go theorem for a quantum field evolved and renormalized on a dynamical compact spacetime; galactic-scale matching, Hubble inference, and a complete compact cosmology are separate calculations. The registered sensitivity suite remains `FAIL` (18/19); qualitative classifications persist in the re-tested cases, but threshold amplitudes are not certified precision predictions.

Keywords: Casimir effect; cosmic topology; Bianchi cosmology; anisotropic expansion; compact manifolds

I. INTRODUCTION

Compact flat three-manifolds admit mode spectra that generically produce direction-dependent vacuum expectation values of the stress tensor [1–3]. That fact is distinct from any claim that a fixed numerical ratio of directional Hubble rates is a free parameter-free prediction of topology.

For definiteness, we use $H_t/H_p = 13/12$ solely as a historical null target for the closure test. Neither a cycle-counting premise nor an observational Hubble value is inserted into the equations. We isolate two controlled questions that can be addressed with the declared lattice formula and biaxial closure integrator (gate package CBR-001):

1. Does the implemented periodic-scalar stress pass its internal numerical and thermodynamic controls and depend on the shape moduli?

2. When that static stress is transported by the declared instantaneous closure on biaxial expansion, does $H_t/H_p = 13/12$ appear as a free-field attractor, a quasi-plateau, or only as a transient crossing (if at all)?

The answers are: the implemented nonzero-mode static stress passes the internal identity checks, agrees with an independent evaluator at relative order 10^{-6}, and is shape dependent; and the bounded closure calculation yields only transient crossings of 13/12, with every valid threshold candidate approaching isotropy through the final integration time $N = 20$.

Shape-dependent Casimir stress itself is not new: rectangular-cavity and periodic mode sums are established [3]. Negro et al. [4] derive topology-dependent scalar stress on de Sitter, discuss the state-dependent isotropic zero mode, and calculate Casimir-sourced anisotropic backreaction. The present contribution is narrower: a reproducible static implementation with local cross-method controls and a finite-domain target-persistence audit of an explicitly declared instantaneous closure. It is neither an independent discovery of anisotropic Casimir stress nor a replacement for their curved-spacetime state calculation.

Companion work on present-epoch scale matching and weak-field matching invariants [5] records independent no-go results for a different geometric construction. The Casimir calculation below is logically independent of that construction.

II. RECTANGULAR LATTICE CASIMIR STRESS

A. Mode sum and pressures

For a massless scalar with periodic lengths (L_1, L_2, L_3), CBR-001 takes as input the already-renormalized

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† GitHub: `github.com/brendohxd/ITSM-...`
Zenodo: doi:10.5281/zenodo.18808348
Gate: `Analysis/Casimir/CBR-001/`

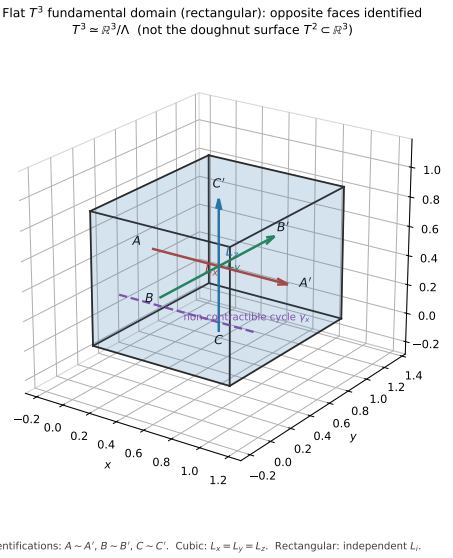


FIG. 1. Rectangular flat T^3 fundamental domain with independent edge lengths L_p and L_t (biaxial case $L_2 = L_3 = L_t$). This is the geometry of the lattice sum, not a Planck-safe cubic cosmology and not an embedded solid torus.

rectangular-periodic Epstein-lattice expression developed in Ref. [3]:

$$\rho_{\rm Cas} = -\frac{\hbar c}{2\pi^2} \sum_{\mathbf{n}\in\mathbb{Z}^3\setminus\{0\}} \frac{1}{\left[(n_1L_1)^2 + (n_2L_2)^2 + (n_3L_3)^2\right]^2}. \tag{1}$$

Directional pressures follow thermodynamically,

$$p_i = -\frac{1}{L_jL_k}\frac{\partial E_{\rm Cas}}{\partial L_i}, \qquad E_{\rm Cas} = L_1L_2L_3\rho_{\rm Cas}. \tag{2}$$

In code and tables we report dimensionless lattice units with the overall $\hbar c$ factor restored only when matching cosmological amplitudes. Figure 1 recalls the rectangular fundamental domain (not an embedded doughnut T^2).

B. Stage-1 validation (cubic point)

At the cube $L_1 = L_2 = L_3 = 1$, the direct cutoff sequence at quadratic extrapolation order gives $\rho = -0.837535893$, while an independent heat-kernel/Mellin evaluation gives $\rho = -0.837536911$. Their relative discrepancy is 1.22×10^{-6}. We therefore quote

$$\rho = -0.837537, \qquad p_i = \rho/3 = -0.279179 \tag{3}$$

in lattice units rather than retaining unsupported extra digits. The direct run has conformal-trace residual $|\rho - \sum_i p_i| \sim 10^{-15}$ at $N = 120$ modes per edge, finite-difference pressure errors $\sim 5\times10^{-10}$, and cubic isotropy at machine precision (Table I). The Stage-1 package reports `STATUS: PASS`.

TABLE I. Cubic Stage-1 diagnostics (lattice units, $L_i = 1$). Extrapolated ρ uses a polynomial retained through $1/N^2$, including a fitted $1/N$ term; validation residuals are at mode cutoff $N = 120$.

Quantity	Value
ρ (direct, order 2)	-0.837535893
ρ (heat kernel/Mellin)	-0.837536911
p_i (quoted cube value)	-0.279179
max FD pressure rel. error	4.8×10^{-10}
conformal-trace residual	$\sim 10^{-15}$
cube isotropy error	$\sim 10^{-16}$

TABLE II. Independent heat-kernel/Mellin benchmark. The final column is the relative energy-density discrepancy.

(L_1, L_2, L_3)	$\rho_{\rm HK}$	$\rho_{\rm dir}$	rel.
$(1, 1, 1)$	-0.8375369	-0.8375359	1.22×10^{-6}
$(1, 1.25, 1.25)$	-0.4788673	-0.4788668	1.20×10^{-6}
$(1, 1, 2)$	-0.4362267	-0.4362262	1.08×10^{-6}
$(0.8, 1.1, 1.4)$	-0.7519731	-0.7519723	1.13×10^{-6}

C. Independent noncubic benchmark and sensitivity

The heat-kernel/Mellin implementation does not call the direct lattice-sum routine. It evaluates the Epstein sum through a Jacobi-theta product and obtains pressures by differentiating the total energy. Table II compares it with the registered direct extrapolation. The maximum normalized pressure discrepancy over these points is 7.42×10^{-7}, while relative trace residuals are below 5.7×10^{-12}. This verifies the nonzero-mode topological stress, not the state-dependent isotropic zero mode retained in a full compact-space treatment [4].

A preregistered release-sensitivity suite varied the cube extrapolation order and fitting window and re-ran Stage 3B at twice the temporal sampling with rtol $= 10^{-12}$ and atol $= 10^{-16}$. All four static precision checks passed. All three registered dynamic shapes retained the `TRANSIENT_CROSSING` classification and no attractor appeared. One threshold amplitude shifted by 8.74×10^{-5} relative to its baseline, however, exceeding the deliberately stringent preregistered 10^{-8} bound; the suite therefore records `FAIL` (18/19), not a rewritten pass. The qualitative classification is stable in the three re-tested shapes, but threshold amplitudes are not quoted as precision predictions. The discrepant pair at $r_0 = 1.5$ is 0.6757299265902508 and 0.6756708783193345. Their difference is not a rigorous uncertainty bound. The threshold objective takes a maximum on sampled times; a certified precision result would additionally require peak-time and sampling refinement, not just a smaller ODE tolerance.

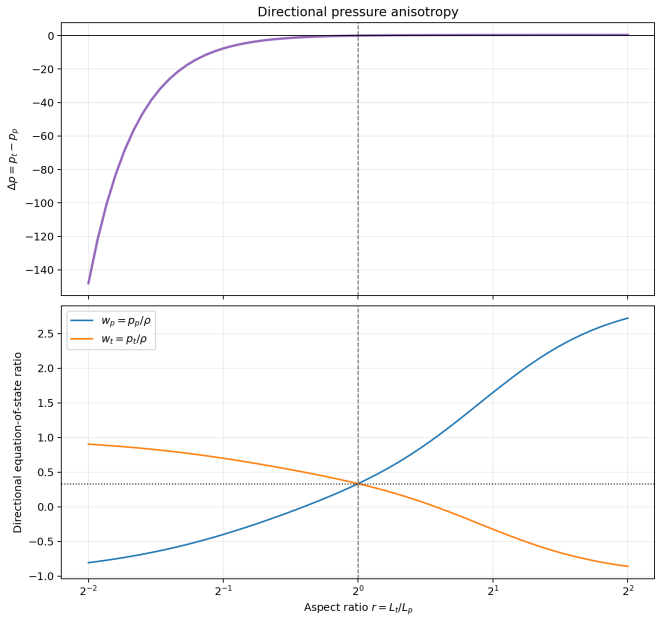


FIG. 2. Stage-2 biaxial anisotropy $\Delta p = p_t - p_p$ versus shape ratio $r = L_t/L_p$. The cubic point $r = 1$ is isotropic; anisotropy is shape moduli dependent.

D. Stage-2 biaxial shape scan

Let $r = L_t/L_p$. CBR-001 Stage 2 holds $L_p = 1$ while varying $L_t = rL_p$; it does not hold the mean scale fixed. Stage 3 converts this raw table to fixed mean scale using the homogeneous $r^{8/3}$ stress rescaling before constructing its source. The stress $\rho(r)$, $p_p(r)$, $p_t(r)$ and the anisotropy $\Delta p \equiv p_t - p_p$ are smooth functions of shape; Δp changes sign across the cube $r = 1$ (Fig. 2). Representative points (lattice units) include

$$r = 1: \qquad \rho = -0.8375, \qquad \Delta p \simeq 0, \tag{4}$$

$$r \simeq 1.26: \qquad \rho = -0.4702, \qquad \Delta p \simeq +0.247, \tag{5}$$

where the second line is in the raw fixed-L_p Stage-2 convention. Compact *shape-dependent* anisotropic stress is therefore a concrete control of the implemented static formula. It is *not* a universal cycle-counting coefficient independent of the mode sum.

III. BIAXIAL INSTANTANEOUS-STRESS CLOSURE

A. Geometry and equations

For the biaxial Bianchi-I line element

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a_p^2\,\mathrm{d}x^2 + a_t^2(\mathrm{d}y^2 + \mathrm{d}z^2), \tag{6}$$

define

$$H = \frac{H_p + 2H_t}{3}, \qquad \delta = H_t - H_p, \qquad r = \frac{a_t}{a_p}. \tag{7}$$

The shear system is the biaxial specialization of Bianchi-I [6], with $\kappa = 8\pi G$,

$$\dot\delta + 3H\delta = \kappa(p_t - p_p), \qquad \dot r = r\delta, \tag{8}$$

and Hamiltonian constraint

$$3H^2 - \frac{\delta^2}{3} = \kappa\rho_{\mathrm{total}}. \tag{9}$$

Stage 3A treats the verified Casimir tensor as a small radiation-like perturbation ($\propto a^{-4}$) on a positive de Sitter background and integrates in $N = \ln a$. It does not evolve quantum-field mode functions on the time-dependent Bianchi-I geometry or renormalize $\langle T_{\mu\nu}\rangle$ there. The dimensionless source-scale ratio ϵ absorbs $\hbar c/L_\star^4$ relative to the de Sitter density scale (CBR-001 Stage 3A convention). No present ITSM derivation fixes this ratio. It checks the perturbative limit and recorded constraint/continuity residuals but does *not* by itself establish any preferred Hubble ratio.

Stage 3B instead retains finite source amplitude and evolving shape. It integrates $(\ln r, \delta)$, using the positive branch

$$H = \sqrt{\frac{\kappa\rho_{\mathrm{total}} + \delta^2/3}{3}}, \qquad \rho_{\mathrm{total}} = \rho_{\mathrm{BG}} + \epsilon\rho_{\mathrm{Cas}}, \tag{10}$$

with constant $\rho_{\mathrm{BG}} > 0$. Thus the mean H responds to the source and shear, rather than being fixed at its de Sitter value. Runs terminate on loss of the supported shape interval, positive total density, real expanding branch, or positive directional expansion. A small residual of Eq. (9), when H is reconstructed from that same equation, checks algebraic consistency, not an independent mean Einstein equation. A separately evolved Raychaudhuri check has not been demonstrated by the recorded controls. This is a constitutive-closure test, not full semiclassical Einstein evolution.

B. Analytic persistence condition

Let

$$x = \frac{\delta}{H}, \qquad q = \frac{H_t}{H_p} = \frac{1 + x/3}{1 - 2x/3}. \tag{11}$$

The historical target $q = 13/12$ corresponds to $x_\star = 3/38$. Because $\mathrm{d}\ln r/\mathrm{d}N = x$, a constant $q \neq 1$ entails $r(N) = r(N_0)\exp[x(N - N_0)]$. It cannot remain in a compact shape interval $0 < r_{\min} \le r \le r_{\max} < \infty$ for arbitrarily large N. This is a kinematic obstruction to a persistent nonunit ratio with bounded shape, independent of a stress prescription; it is not a fixed-torus property. For a prescribed $x(N)$, Eq. (8) requires the anisotropic pressure

$$\Pi_{\mathrm{req}} = \frac{H^2}{\kappa}\left[\frac{\mathrm{d}x}{\mathrm{d}N} + x\left(3 + \tfrac{\mathrm{d}\ln H}{\mathrm{d}N}\right)\right]. \tag{12}$$

In de Sitter at fixed $x = 3/38$, this becomes $\Pi_{\rm req} = 9H^2/(38\kappa)$. By contrast, the free Casimir anisotropy in the declared closure scales as

$$\Pi_{\rm Cas} \propto a^{-4}\, r^{8/3}\big[\widehat{p}_t(r) - \widehat{p}_p(r)\big]. \tag{13}$$

For unbounded expansion $a \to \infty$ and shape in the compact interval above, the bounded shape factor makes $\Pi_{\rm Cas} \to 0$. If also $H \to H_\infty > 0$, $\mathrm{d}\ln H/\mathrm{d}N \to 0$, $x \to x_\star$, and $\mathrm{d}x/\mathrm{d}N \to 0$, the required stress tends to $9H_\infty^2/(38\kappa) \neq 0$. These stated asymptotics give a second, stress-scaling inconsistency; positive H alone is insufficient. Neither argument covers unbounded shape, a finite total expansion, or an $H \to 0$ branch. Nor do they replace dynamical quantum-state evolution or renormalization. The finite-domain scan below measures transient crossings and quasi-plateaus through $N = 20$; it is not a numerical proof of an infinite-time limit or an all-action no-go theorem.

C. The finite-domain Stage-3B ratio test

The small-source fixed-background profile $x(N) = S(e^{-3N} - e^{-4N})$ peaks at $N = \ln(4/3)$ with maximum $(27/256)S$; merely reaching $x_\star$ already requires $S \geq 128/171 \simeq 0.7485$, outside the strict small-perturbation regime.

A nonlinear root search over initial shapes $r_0 \in \{1.01, 1.05, 1.1, 1.25, 1.5, 2, 3, 4\}$ and source-scale ratios spanning seven decades finds target-reaching thresholds for five of eight shapes. Classification vocabulary:

- `NO_CROSSING`: peak q stays below 13/12;
- `TRANSIENT_CROSSING`: touch or brief crossing;
- `QUASI_PLATEAU`: dwell ≥ 1 e-fold within 1% without attractor criteria;
- `ATTRACTOR`: dwell ≥ 5 e-folds within 1%, nearby trajectories converge, and no subsequent decay toward $q = 1$.

Results: zero attractors, zero quasi-plateaus, five transient crossings, two no-crossings, and one trajectory initialized at the upper interpolation boundary whose outward drive leaves the supported domain (Table III; Fig. 3). Dwell times within 1% of the target are only ~ 0.27–0.30 e-folds. Every valid threshold candidate approaches $q = 1$ through $N = 20$, consistent with the declared a^{-4} source becoming negligible. The tested nearby trajectories also decay toward isotropy; none satisfy the registered attractor criteria.

The source fraction $F_{\rm Cas} = \max_N |\epsilon\rho_{\rm Cas}/\rho_{\rm total}|$ is respectively 2.536, 0.632, 0.273, 0.133, 0.082 for $r_0 = 1.1, 1.25, 1.5, 2, 3$ at the recorded thresholds. The denominator includes the negative Casimir density, so the fraction can exceed unity. Four cases exceed the registered nonperturbative threshold $F_{\rm Cas} \geq 0.1$; the last is marginal ($0.01 \leq F_{\rm Cas} < 0.1$). These are finite-amplitude diagnostics of the declared closure, not controlled small-source or renormalized quantum-backreaction predictions.

TABLE III. Stage-3B classification at threshold amplitudes for the 13/12 search. "Regime" labels the peak Casimir energy fraction (perturbative / marginal / nonperturbative). Rounded thresholds are diagnostic locations, not precision predictions or certified error bars.

r_0	class	$\epsilon_\star$	regime
1.01	no crossing	—	—
1.05	no crossing	—	—
1.10	transient	2.55	nonpert.
1.25	transient	1.34	nonpert.
1.50	transient	0.676	nonpert.
2.00	transient	0.278	nonpert.
3.00	transient	0.089	marginal
4.00	invalid	—	—

IV. SCOPE AND OPEN ROUTES

- Free massless scalar only; interacting or multi-field sectors untested.
- The renormalized static lattice expression is supplied as input. A local independent heat-kernel/Mellin evaluation now agrees at four points, but an external replication and a full-domain error analysis remain outstanding.
- The time-dependent calculation is an instantaneous a^{-4} closure, not a mode-function evolution or curved-spacetime renormalization of $\langle T_{\mu\nu}\rangle$. Particle production, curvature counterterms, trace-anomaly and memory terms are not calculated.
- Rectangular moduli; twisted flat spaces (E_2, E_3) not scanned.
- Stage 3A is perturbative on de Sitter; Stage 3B permits finite amplitude and constraint-reconstructed H. Four target-reaching thresholds are nonperturbative and one marginal. No controlled quantum-state approximation has been established for those finite-amplitude histories.
- The preregistered sensitivity suite is complete but not all-pass: one threshold-amplitude stability check missed its 10^{-8} bound, although all re-tested classifications and the bounded no-attractor conclusion were unchanged.
- A derived driven or memory stress (CBR-002) is one possible extension, not a demonstrated solution or an exhaustive classification of other actions.

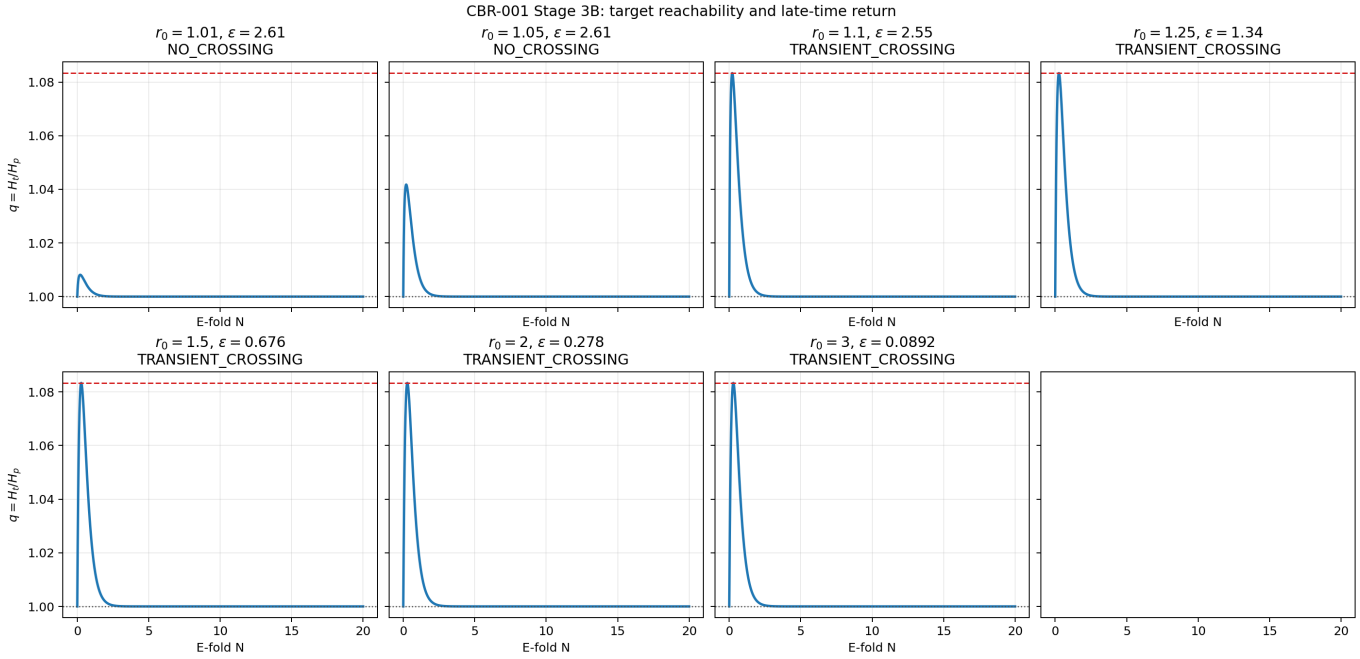


FIG. 3. Representative Stage-3B trajectories of $q = H_t/H_p$ versus N. The horizontal line marks 13/12. Passages, when present, are brief; late evolution approaches isotropy over the displayed finite interval. The plotted data and their original hashes are unchanged.

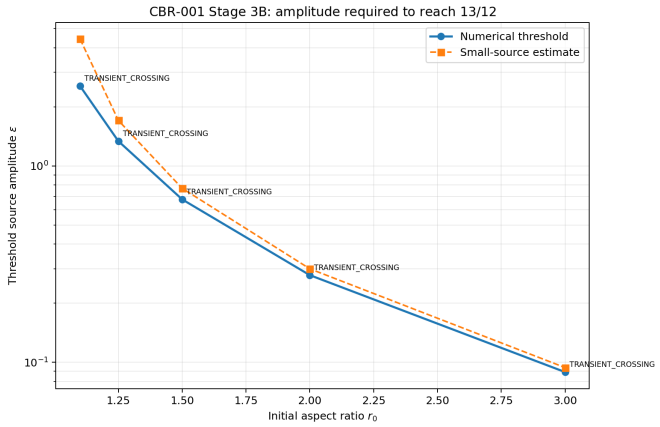


FIG. 4. Stage-3B threshold amplitudes $\epsilon_\star(r_0)$ for first touch of $q = 13/12$. Near-cubic shapes require large source-scale ratios; no threshold defines an attractor.

- CMB compatibility of any particular (L_i) is not established here.

These assumptions delimit the result. Driven constitutive stress, alternate topologies, richer field content, and dynamical renormalization remain open routes that require their own conservation, stability, and observational tests.

V. CONCLUSIONS

A free massless scalar on rectangular flat T^3 yields an internally checked and independently benchmarked nonzero-mode, shape-dependent anisotropic Casimir stress. When that static stress is transported as an instantaneous a^{-4} source in the finite-amplitude biaxial closure, the ratio $H_t/H_p = 13/12$ appears at most as a brief transient at large source-scale ratios, never as a quasi-plateau or attractor in the bounded search reported here. The tested closure therefore cannot sustain the historically proposed anisotropic expansion or justify a map to $H_t = 72.97\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The bounded negative result has a constructive consequence: it separates a verified static, shape-dependent stress from the additional constitutive physics needed to maintain anisotropic expansion. A persistent nonunit ratio cannot stay in a compact shape interval over unbounded expansion; the additional stress-scaling obstruction assumes $H \to H_\infty > 0$ and the regular asymptotics stated in Sec. III. The numerical negative result concerns only the registered finite-domain search. It does not exclude unbounded-shape or $H \to 0$ branches, nor a derived driven, nonlocal, interacting, or multi-field response. The next test, CBR-002, must derive such a stress independently, verify conservation and stability, and only then compare its evolution with the target ratio without inserting that ratio into a coefficient.

CODE AND DATA AVAILABILITY

The companion codebase is maintained in the public GitHub repository. The local review package identifies the Stage-1–3B implementation, frozen CSV/JSON outputs, local cross-method controls, and preregistered sensitivity contract and receipt. An immutable public snapshot of this exact revised version must be identified before publication; this draft does not assert that its new review package has already been released. SHA-256 anchors identify exact artifacts; scientific validity rests on the declared equations, assumptions, and tests.

VERIFICATION STATUS

The preregistered release-sensitivity suite currently passes 18 of 19 checks. The sole failure is a threshold-amplitude repeatability test at $r_0 = 1.5$: the relative difference is 8.74×10^{-5}, above the registered 10^{-8} bound. Classifications of the three re-tested shapes and the bounded no-attractor conclusion are unchanged. This release candidate remains on publication hold pending deferred external review and explicit operator approval; artifact hashes are provenance records, not scientific clearance.

DECLARATION ON THE USE OF ARTIFICIAL-INTELLIGENCE TOOLS

Artificial-intelligence-assisted tools were used for editorial drafting, source-code assistance, and checks of symbolic and numerical consistency. This local review draft still requires the author's final verification of the manuscript, equations, cited sources, and generated artifacts. Submission requires the author's confirmation of that review and acceptance of full responsibility; neither is certified by these automated checks.

ACKNOWLEDGMENTS

Computations use the repository environment `itsm_env` and the CBR-001 gate package. Claim hygiene for related geometric packaging is recorded in companion note [5].

Appendix A: Reproducibility anchors

Table IV lists SHA-256 prefixes for portable CBR-001 CSV/JSON artifacts underlying the Stage-1–3B numbers in the main text (branch `recovery/v12-core-architecture`). CSV writers use a fixed LF record terminator and JSON output paths are relative to the summary directory. Re-running the Stage scripts after intentional code changes will alter digests; any published revision must update this table.

TABLE IV. SHA-256 of CBR-001 artifacts used in this draft.

Artifact	SHA-256 (prefix)
`cbr001_stage1.csv`	`57847F382BAF...`
`cbr001_stage2_scan.csv`	`FE648629545F...`
`cbr001_stage3b_thresholds.csv`	`94B0AE4C6381...`
`cbr001_stage3b_summary.json`	`9A5E80D2F713...`

Full digests and reproduction commands appear in the companion `CBR001_CHECKSUMS.md` ledger.

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